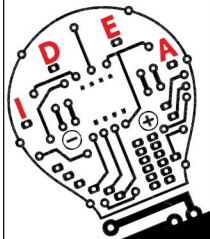


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Fig. 4: PCB section of the software

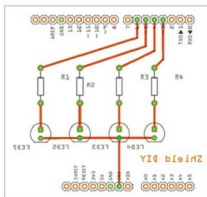


Fig. 5: Arduino shield single-side PCB



Fig. 6: Choose Export for PCB option after proper routing

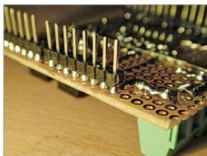


Fig. 7: Berg strip soldering on track side of the PCB

designing of schematics, the option of Breadboard is recommended. The schematic made using Fritzing is shown in Fig. 3.

4. Go to PCB section of Fritzing as shown in Fig. 4. Select the type of PCB you need to make. Various options are available like Auto Routing and Layers Selection.

5. Select the settings for Copper Layer as per requirement. Route connections manually as per the circuit and the requirement. Fig. 5 shows

Arduino shield PCB after routing. The settings are different for double- and single-side PCBs.

6. After preparing the PCB and proper routing of the circuit, click Export for PCB as shown in Fig. 6 and save the files in PDF format in the destination folder.

7. Check the design rules by clicking Routing under PCB section.

8. If copper fill is required, it can be applied to the PCB using the relevant option under Routing options.

9. Place the berg strips/headers used for connecting the shield to Arduino UNO board on the same side for soldering on a single-side PCB. Careful soldering is required for this purpose. Since the berg strip header and routing tracks are on the same side, remove the plastic sheath from the berg strip before soldering. Place the sheath back after soldering. A long header of 25mm is recommended. Placement of the same is shown in Fig. 7.

10. Specify the type of file in which you are exporting the circuit diagram PCB artwork. If you export in PDF format, specify the folder where you want to save the exported file. A number of files will be saved in the target folder in PDF format for bottom-layer copper, top-layer copper, mask top, mask bottom, silk top, silk bottom and so on, and their respective mirror image files will be saved in another PDF file. We can easily select the desired file for a particular requirement.

11. Transfer the artwork on the PCB, which is cut to the required size. Proceed for etching and soldering of components.

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EFY Note

The relevant files of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com



Madhuran Mishra is a student of ME digital communication at NITTR, Bhopal